16TH INTERNATIONAL PROJECT COMPETITION AND EXHIBITION

24.02.2026

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SDG/THEME CHOOSEN: **Industry , Innovation and Infrastructure**

Title of the Project: **Design And Development Of Smart Fire Extinguisher Using LoRa**

College Name :Vel Tech Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology

Reference Number :III260084

Index

Project details

OFFLINE COMMS

- LoRa (433MHz)
- No Wi-Fi Needed
- 2km Range

SMART DETECTION

- Triple-Check Logic
- Gas + Heat + Flame
- 0% False Alarms

AUTO-ACTION

- Servo Actuation
- Instant Trigger
- Mechanical Press

Concept of the solution

- Transmitter-Receiver design
- Multi-sensor surveillance with ESP32
- Three states of the system: SAFE / TEMP_GAS / FLAME
- Automatic suppression with servo

Novelty / Scope of Solution

Novelty:

- Multi-state fire detection
- Priority-dependent flame reaction
- Adaptive LoRa transmission periods
- Fire duration and event recording

Scope:

- Industrial settings
- Warehouses
- Chemical storage
- Remote locations

Technical DescriptionSystem Architecture:

- ESP32-based controller
- DHT22 + Gas + Flame sensors
- Servo motor for extinguisher
- LoRa communication module

Working:

- Flame → Servo 180° (Activate)
- Temp/Gas high → Warning alert
- State-change based messaging

Implementation Plan/Working Model

- Continuous sensor monitoring
- Threshold: Temp > 45°C, Gas > 1800
- LoRa data transmission
- Receiver logs fire events

Explanation of Demonstration:

Working model was demonstrated through video presentation
video Link

Validation / Testing / Analysis

- Sensors tested under controlled fire conditions
- Servo activation verified
- LoRa communication validated
- Fire duration recorded correctly

Cost Estimate(if required)

Component Category	Component Name	Qty	Unit Price (₹)	Total Cost (₹)	Role in Project
Microcontrollers	**ESP32-S SoC Board**	1	550	550	**Sender Node** (Detector) Brain
	ESP32 WROOM 32 DevKit	1	580	580	**Receiver Node** (Dashboard) Brain
Communication	LoRa Module (Ra-02 / XL32 433MHz)	2	380	760	Long-range offline communication
Sensors	MQ-135 Gas Sensor	1	160	160	Smoke/Gas detection
	DHT22 Temp & Humidity Sensor	1	280	280	Heat detection (Precision)
	IR Flame Sensor	1	90	90	Instant fire flame detection
Security & Input	RFID-RC522 Module (Reader + Card)	1	180	180	Auth for System Reset
	Push Button (Tactile 12x12mm)	**1**	**10**	**10**	**Manual Reset / Test Trigger**
Actuators & Display	MG996R Metal Gear Servo	1	380	380	Mechanical Arm (High Torque)

Pros and Cons of the solution

Advantages

- Fully automatic response
- long range alerting
- Real-time monitoring

Disadvantages

- Needs mainatenance
- Refill of extinguisher required

References

S.No	Title	Source	Year	Working Link
1	LoRaWAN Network for Fire Monitoring in Rural Environments	Electronics (MDPI)	2020	Link
2	Forest Fire Monitoring and Energy Optimization Using LoRa Mesh	Electronics (MDPI)	2025	Link
3	A Fault-Tolerant Surveillance System for Fire Detection Using LoRaWAN	Sensors (MDPI)	2022	Link
4	LPWAN Based IoT Surveillance System for Outdoor Fire Detection	*IEEE Access* (Q1)	2020	Link

THANK YOU